



Design & Deployment of a Secure Cloud-Based PDMS

Using a Three-Tier Architecture on AWS

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Chapter 1: Introduction

Background of the Study

-  **Rapid Digital Transformation:** Transition from paper-based to Electronic Health Records (EHR) in clinics.
-  **Sensitive Data Protection:** Healthcare Information Management requires robust infrastructure to guard PHI.
-  **Rising Cyber Threats:** Growing complexity of attacks targeting small healthcare facilities.
-  **Legacy Inefficiency:** Traditional on-premises solutions lack the scalability and security of cloud paradigms.



Problem Statement



Flat Architecture

On-premises systems often lack network segmentation, allowing malware to move laterally across tiers easily.



Manual Deployment

Configuration via FTP/USB drives prevents automated security gates and quality checks before production.



Ransomware Victim

Case study: Alamin Clinic suffered a data-encrypting breach due to these legacy vulnerabilities.

Project Aim and Objectives

AIM: To design and deploy a secure cloud-based PDMS for Alamin Clinic using a three-tier architecture on AWS with DevSecOps integration.

-  To investigate core concepts: Cloud security, 3-tier VPC architecture, and IAM practices in healthcare.
-  To develop a secure PDMS: Covering VPC networking, RBAC through IAM, and a DevSecOps CI/CD pipeline.
-  To evaluate system performance: Using vulnerability scans (Trivy, SonarQube) and RTO stress testing.
-  To validate compliance: Assessing HIPAA posture using AWS Security Hub.

Project Scope Boundaries

✓ In-Scope

- VPC Public/Private Subnets
- React/Node.js/RDS (Three-Tier)
- IAM Least Privilege & RBAC
- KMS AES-256 Encryption
- Terraform (IaC) & GitHub Actions

✗ Out-of-Scope

- Hospital Billing & Insurance Claims
- Pharmacy Inventory Management
- Emergency Room (ER) Ops
- Specialized Clinical Modules (O&G)

System Resilience Significance

15m

Recovery Time
Objective

Technical & Clinical Impact

The proposed system transforms the recovery process from a **5-day manual restoration** (prone to data loss) into a **sub-15 minute automated rebuild** using Infrastructure as Code (Terraform).

This ensures high availability for doctors and prevents the operational "blackouts" seen during the Alamin Clinic ransomware incident.

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Chapter 2: Literature Review

Market Gap Analysis

System Type	Security Model	Deployment	Ransomware Risk
Legacy On-Premise	Reactive / Perimeter	Manual FTP / USB	High (Single Point)
OpenEMR (Open Source)	Community Patches	Variable / Scripted	Moderate
Epic Systems	Proactive / Managed	Enterprise SaaS	Low (High Cost)
Proposed PDMS	Shift-Left / DevSecOps	Automated (IaC)	Extremely Low

Case Study: Alamin Clinic

Anatomy of an Attack

A private Saudi Arabian healthcare provider faced a total operational halt when ransomware encrypted their flat-network servers.

Key Vulnerabilities:

- No network isolation between DB and Web.
- Manual, reactive antivirus updates.
- 72-hour blackout + 5-day recovery window.

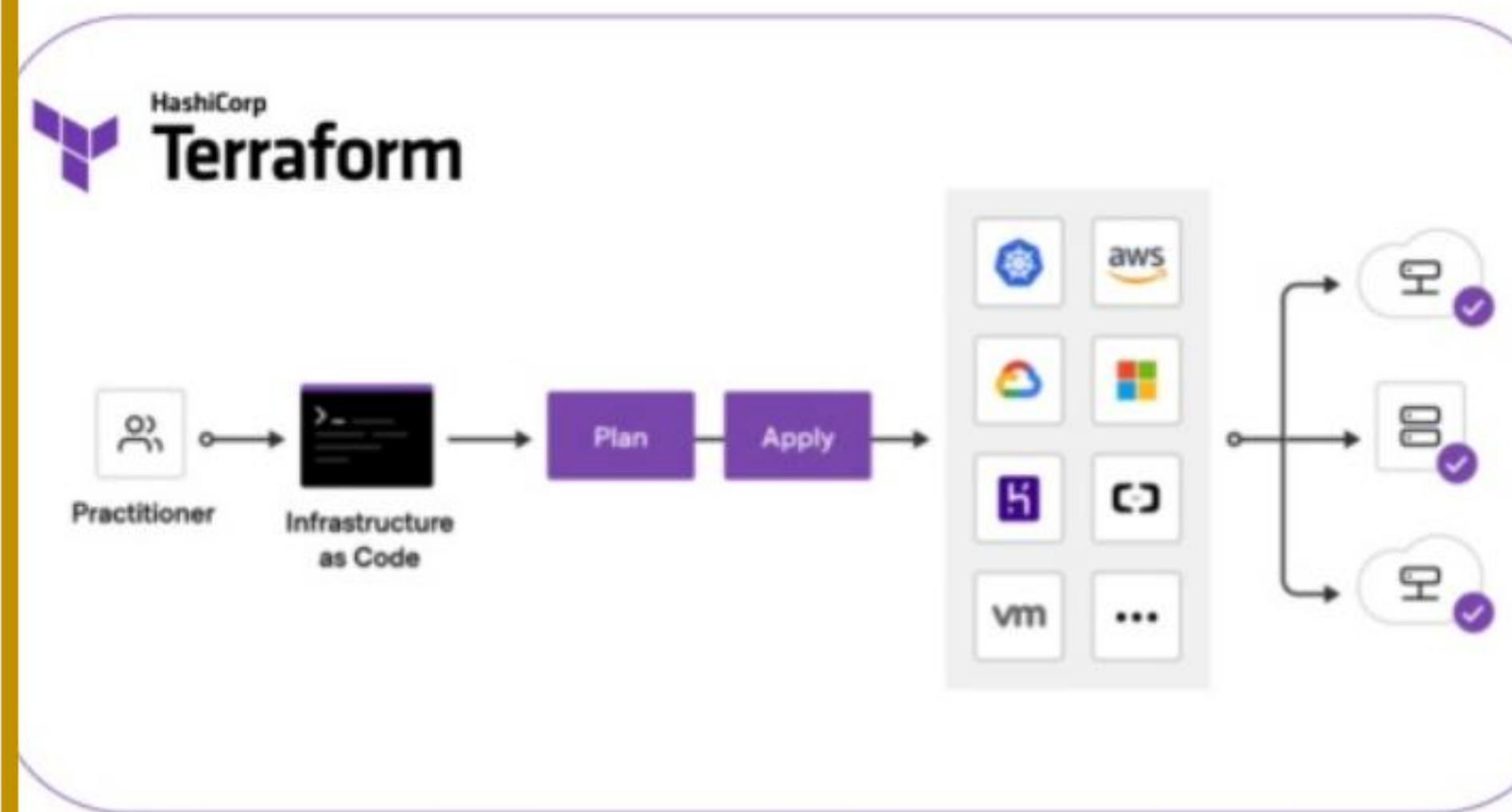


Core Technologies & Infrastructure

 AWS 3-Tier Architecture

AWS VPC

Three-tier architecture isolating Web, App, and RDS layers.



Terraform IaC

Declarative configuration for reproducible infrastructure.



DevSecOps

Automated security gates: Trivy, SonarQube, and Checkov.

Recovery Time Performance

Legacy Manual Restore

120 Hours

Proposed IaC Recovery

25
Hours

Switching to Infrastructure as Code reduces recovery time by 99.8%, ensuring patient care is not interrupted by extended outages.

Questions?

Thank you for your attention.

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Image Sources



<https://upload.wikimedia.org/wikipedia/commons/8/81/UTM-LOGO.png>

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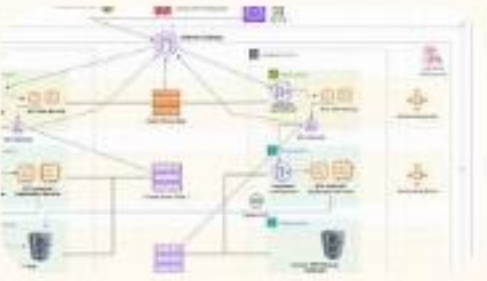
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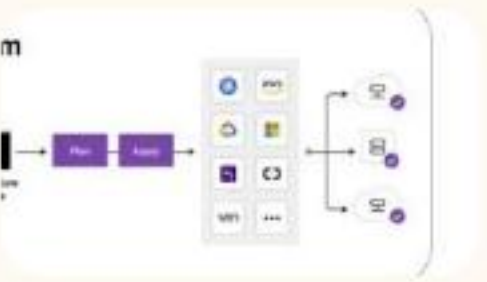
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